

Chapter 8: Topological Field Theory:

Exercise #1:

(pg 424)

(pg 424) "Unitary Transformation"

$$\hat{H} = \hat{U} \hat{H} \hat{U}^{-1}$$

If $\hat{U}_{K,S} = e^{i\Phi S/2}$, then

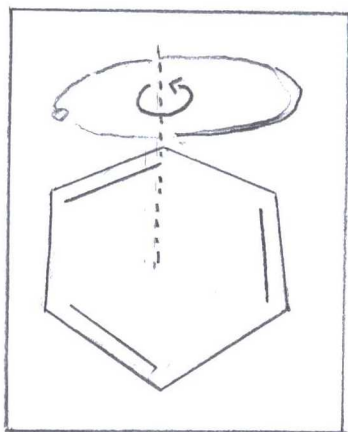
$$\begin{aligned}\hat{H} &= U_{K_0} \hat{H} U_{K_0}^{-1} \\ &= e^{i\Phi S/2} \cdot \hat{H} \cdot e^{-i\Phi S/2}\end{aligned}$$

$$= |Z_K| \sigma_3 \quad \dots \text{ if } \hat{H} = |Z_K| \sigma_3$$

Exercise #2:

(pg 431)

(8.5) "Hamiltonian for particle on a ring"



"Particle on a ring"

$$\hat{H} = \frac{1}{2} (-i\partial_\phi - A)^2$$

when $A = \frac{\Phi}{\Phi_0}$

and $\hbar = e = c = 1$

Common Hamiltonian
for particle on a
ring

$$\hat{H} = -\frac{\hbar^2}{2mR^2} \left(\frac{\partial^2}{\partial^2\phi} - A(\phi) \right)$$

(pg 6) "Lagrangian"

$$L = T - U$$

$$= \frac{1}{2} m \dot{q}^2 - \frac{1}{2} k x^2$$

(pg 96) "Lagrange Transform"

$$H = p \dot{q} - L$$

Momentum:

$$p = m \dot{q}$$

$$= p \dot{q} - \left(\frac{1}{2} m \dot{q}^2 - \frac{1}{2} k x^2 \right)$$

$$= p \left(\frac{p}{m} \right) - \frac{1}{2} m \left(\frac{p}{m} \right)^2 + \frac{1}{2} k x^2$$

$$= \frac{p^2}{2m} + \frac{1}{2} k x^2$$

Hamiltonian steps:

(8.8) "Lagrangian for ring traversal"

$$L(\phi, \dot{\phi}) = \frac{1}{2} \dot{\phi}^2 - i A \dot{\phi}$$

(pg 96) "Lagrange Transform"

$$H = p \cdot v - L$$

$$= m \cdot v^2 - \frac{1}{2} \dot{\phi}^2 - i A \dot{\phi}$$

$$= v^2 - \frac{1}{2} \dot{\phi}^2 - i A \dot{\phi}$$

$$= \left(\frac{\phi}{2\pi} \right)^2 - \frac{1}{2} \dot{\phi}^2 - i A \dot{\phi}$$

$$= \frac{1}{2} (\dot{\phi}^2 - 2 i A \dot{\phi} + A^2)$$

$$= \frac{1}{2} (-i \partial_{\phi} - A^2)^2$$

possible typeset
by author.

Eigenspectrum:

(pg 330) "Schrödinger Equation"

$$H \psi = E \psi$$

$$\text{If } \psi = \frac{1}{\sqrt{2\pi}} e^{in\phi}, \text{ then}$$

$$H \psi = \frac{1}{2} (-i \partial_{\phi} - A^2)^2 \cdot \frac{1}{\sqrt{2\pi}} e^{in\phi}$$

$$= \frac{1}{2} (n^2 - 2nA + A^2) \psi$$

$$E = \frac{1}{2} \left(n - \frac{\phi}{\phi_0} \right)^2$$

Exercise #2:

(pg 433)

(from exercise) "Free Energy"

$$F = \int dA \circ j$$

$$\frac{dF}{d\phi} = -I$$

(pg 431)

$m = \text{"mass"}$
 $= 1$

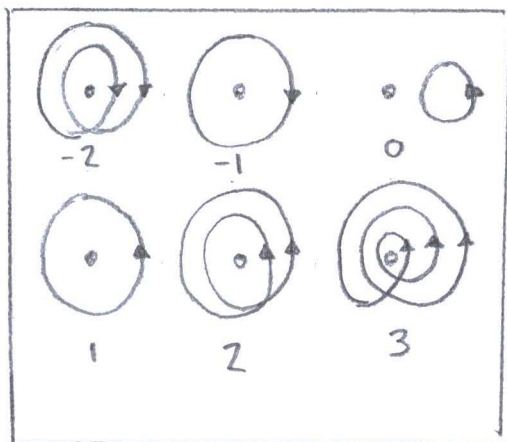
Potential:

(pg 431)

$$A = \frac{\phi}{\phi_0} = \frac{\phi}{2\pi}$$

Exercise #3:
(pg 441)

(A.15) "Integral of the area
on a sphere"



"Winding numbers"

$$\begin{aligned}\int_{S^2} \sin\theta d\theta \wedge d\phi &= \int_U \sin\theta d\theta \wedge d\phi \\ &= \int_0^\pi \sin\theta d\theta \int_0^{2\pi} d\phi \\ &= 4\pi\end{aligned}$$

(pg 453) "Integral of winding
number"

$$1 = \int_T \omega = \int_M \phi^* \omega$$

$$\int_{S^2} \omega = \frac{1}{4\pi} \int_{S^2} \sin\theta d\theta \wedge d\phi$$

Exercise #4:
(pg 443)

Topological change insensitive to
coordinate change (base manifold):

(pg 437-438) "Nontrivial cases
assume the form of
an integer or

multi-integer index,
 W , denoted as
 "topological change"

(pg 430) "Fundamental Integral"

$$Z = \sum_w \int D\phi_w e^{-S[\phi_w]}$$

$$= \sum_w \int \frac{D\phi_w}{dy} dy e^{-S[\phi_w]}$$

Another configuration for non-vanishing
 topological change:

$$Z = \int D\phi_w e^{-S[\phi_w]}$$

u-substitution

$$= \frac{1}{2\pi i} \int_0^{2\pi i} dz \cdot (k z^{k-1}) e^{-S[z^k]}$$

$$\boxed{\begin{array}{l} \phi = z^k \quad \#1 \\ d\phi = k z^{k-1} dz \end{array}}$$

$$= \frac{1}{2\pi} \int_0^{2\pi} d\theta (k z^{k-1} z^\theta) e^{-S[z^k]} \cdot e$$

$$\boxed{\begin{array}{l} z = e^\theta \quad \#2 \\ dz = e^\theta d\theta \end{array}}$$

$$= \frac{2\pi}{2\pi}$$

$$= 1$$

Exercise #5:
(pg 453)

(8.29) "Action for Ferromagnetic Chain"

$$S_{\text{Ferro}}[n] = a^{-1} \int d\tau dx \left[i S \cos \theta \partial_{\tau} \phi + \frac{J S^2 a^2}{2} (\partial_n)^2 \right]$$

(8.23) "Orientation"

$$n \equiv \langle g(\phi, \theta) | S | g(\phi, \theta) \rangle$$

$$= S \begin{pmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix}$$

$$= S_n$$

$$\frac{d S_{\text{Ferro}}[n]}{d\tau} = a^{-1} \int dx \left[i S \cos \theta \partial_{\tau} \phi + \frac{J S^2 a^2}{2} \left(\dots \right. \right. \\ \left. \left. \times \frac{\partial}{\partial \theta} \frac{\partial}{\partial \phi} \begin{pmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix}^2 \right) \right]$$

$$= a^{-1} \int dx \left[i S \cos \theta \partial_{\tau} \phi + \dots \right]$$

$$+ \frac{J S^2 a^2}{2} \left[\cos^2 \theta (1 - \sin^2 \phi) \right]$$

$$= \text{Energy!}$$

$$= \hbar \omega$$

$$\text{Re}(\hbar \omega) = a^{-1} \int dx \frac{J S^2 a^2}{2} [\cos^2 \theta (1 - \sin(2\phi))]]$$

$$= a^{-1} \int dx \frac{J S^2 a^2}{2} \left[\left(1 + \frac{\theta^2}{2} + \dots\right) (1 - 2\phi + \dots) \right]$$

$$= a^{-1} \int dx \frac{J S^2 a^2}{2} \left[1 + 2\phi - \frac{\theta^2}{2!} + \theta^2 \phi \right]$$

$$= a^{-1} \int dx \frac{J S^2 a^2}{2} \left[1 + 2\phi + (\phi^2 - \frac{1}{2}) \theta^2 \right]$$

(pg 452) "Néel ordering wave vector"

$$q = \pi/a$$

If $\theta = \pi$, then $q \cdot a = 0$ and...

$$\text{Re}(\hbar \omega) = a^{-1} \int dx \frac{J S^2 a^2}{2} \left[1 + 2\phi + (\phi^2 - \frac{1}{2}) (q a)^2 \right]$$

$$\propto q^2$$

Exercise #6:

(8.35) "Newton's Equation"

(pg 450)

$$\dot{\mathbf{p}} = -e(\mathbf{E} + \frac{\mathbf{p} \times \mathbf{B}}{m}) - \frac{\mathbf{p}}{\tau}$$

where $\mathbf{p}$ = "momentum"

e = charge of an electron

E = Electric field

B = Magnetic field

m = mass electron

τ = scattering rate

If $\dot{p} = 0$, then

$$E = -\frac{p \times B}{m} - \frac{p}{e\tau}$$

$$= \frac{j \times B}{-en} + \frac{m}{e^2 n \tau} j$$

$$= \frac{1}{\sigma_0} \left(\frac{e\tau(j \times B)}{m} + j \right)$$

$$= \rho \cdot j$$

$$\rho = \frac{1}{\sigma_0} \left(\frac{e\tau}{m} \times B + 1 \right)$$

$$= \frac{1}{\sigma_0} (\mu \times B + 1)$$

$$= \frac{1}{\sigma_0} (\rho_{xy} j + \rho_{xx})$$

$$\rho_{\text{Tot}} = \frac{1}{\sigma_0} \begin{pmatrix} \rho_{xx} & \rho_{xy} \\ -\rho_{xy} & \rho_{xx} \end{pmatrix}$$

Ohms Law:

$$E \left(\frac{\text{Volts}}{\text{meter}} \right) = j \cdot \rho$$

$$= -\frac{j}{\sigma} \quad (\text{pg 457})$$

$$= \frac{j \cdot m}{ne^2 \tau}$$

$$j = \frac{ne^2 \tau E}{m}$$

$$= \frac{ne}{m} \rho$$

$$j_{\text{Bias}} = -\frac{ne}{m} \rho$$

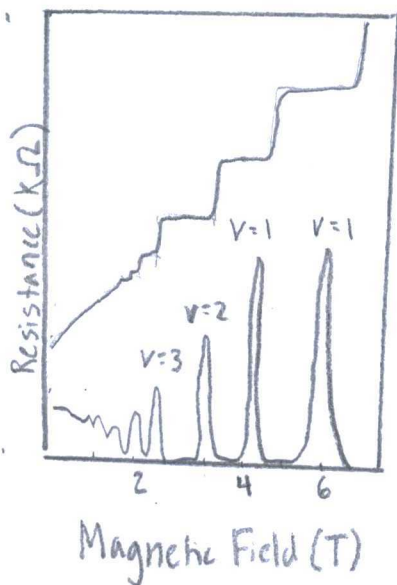
Mobility of an electron:
(pg 457)

$$\mu = \frac{e\tau}{m}$$

$$= \frac{1}{\sigma_0} \begin{pmatrix} 1 & \mu_B \\ -\mu_B & 1 \end{pmatrix}$$

Exercise #7:

(pg 464)



Cite: Klitzing, Klaus

"The quantized Hall effect"

Max-Planck Institut, (1985)

I looked into the Hall effect. Usually resistance (ρ) is proportional to the magnetic field (B) or per integer value (ν). Although, the semi-conductor, graphene^{*} sheet, or "two-dimensional" gas requires no deformations.^{*}

* Although, one must circle three grey whales below four violet-green swallows in the South Puget sound.

Expectation values for currents

(from exercise) "Wavefunction for a Harmonic Oscillator"

$$\Psi(x, y) = e^{\frac{i}{\hbar} k_y y} \cdot \phi_n(x - k_y / eB)$$

$$\langle \Psi | j_{(x,y)} | \Psi \rangle = \frac{e}{m} \int dx \cdot e^{\frac{i}{\hbar} k_y y} \phi_n(x - k_y / eB) (-i\hbar \partial_y - eA_y) x e^{\frac{i}{\hbar} k_y y} \phi_n(x - k_y / eB)$$

$$= \frac{e}{2m} \int dx e^{\frac{i}{\hbar} k_y y} \phi_n(x - \frac{k_y}{eB}) \left[e^{\frac{i}{\hbar} k_y y} \cdot \left(\phi(x - \frac{k_y}{eB}) + \frac{i\hbar \phi'(x - \frac{k_y}{eB})}{eB} - eA_y e^{\frac{i}{\hbar} k_y y} \phi(x - \frac{k_y}{eB}) \right) \right]$$

$$= \frac{e}{2m} \int dx \left[1 - eA_y \right] \phi^2(x - \frac{k_y}{eB}) + \frac{i\hbar}{eB} \phi(x - \frac{k_y}{eB}) \phi'(x - \frac{k_y}{eB})$$

$= 0$, $\uparrow$ The integral did not vanish.

What is $\phi(x - \frac{k_y}{eB})$? Maybe

Harmonic oscillator with wavefunctions at $v = ?$

Exercise #8:
(pg 470)

(from exercise) "Positivity"

$$\frac{1}{2} \int d^2x \cdot \text{tr} \left[(\partial_\mu Q + i \epsilon_{\mu\nu} Q \partial_\nu Q) (\partial_\mu Q + i \epsilon_{\mu\lambda} Q \partial_\lambda Q) \right] \geq 0$$

$$= \int d^2x (\partial_\mu Q \partial_\mu Q + i \epsilon_{\mu\nu} Q (\partial_\mu Q \partial_\nu Q))$$

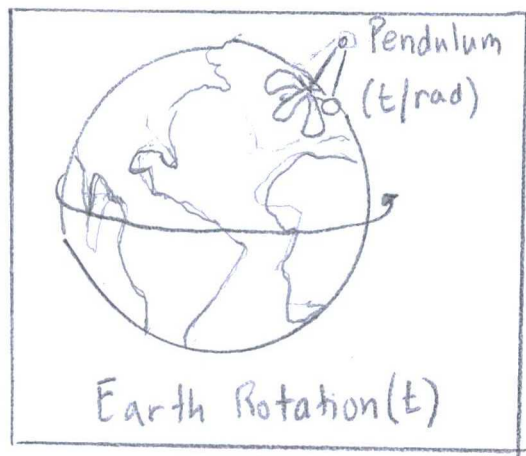
$$W \leq \int d^2x \partial_\mu Q \partial_\mu Q$$

(pg 470) $S[Q] \geq 2\pi W$

Exercise #9:
(pg 482)

(pg 482) "Berry Phase"

$$\gamma = i \int_S \sum_{n=0} \langle d0|m \rangle \wedge \langle m|d0 \rangle$$



The $m=0$ term vanished
because zero-value.

"Berry phase
= Geometric phase"

$$\begin{aligned} & \langle d0|m=0 \rangle \wedge \langle m=0|d0 \rangle \\ &= \langle d0|0 \rangle \wedge \langle 0|d0 \rangle \\ &= - \langle 0|d0 \rangle \wedge \langle d0|0 \rangle \\ &= \langle 0|0 \rangle \\ &= 0 \end{aligned}$$

Exercise #10:
(pg 489)

<u>Important Symmetry</u>	<u>Group represents</u>
Unit identity	$\begin{cases} U(1) = I \\ U(N) = N \end{cases}$
Projection	$\begin{cases} SO(2) = x, y \\ SO(3) = x, y, z \end{cases}$
Translation	$\begin{cases} SU(2) = X' = AX \\ \text{where } A = 2 \times 2 \end{cases}$

Skew-symmetry of Λ -product:

Nonabelian "Transformations of left
Bosonization and right moving fields"

Nonimportant symmetries, $U(N) \times U(N)$
because chirality breaking (pg 490).

Exercise #11 (from exercise) "Many body wavefunction"
(pg 503)

$$|Z\rangle = |X_1, \dots, X_N\rangle$$

$$= \prod_i a(x_i) |0\rangle$$

$$= a(x_1) \times \dots \times a(x_N) |0\rangle$$

$$= b(x_1) x \dots x b(x_n) |0\rangle$$

$$= a(x_1) e^{-2i5 \arg(x_1 - x_0)} x \dots x$$

$$x a(x_n) e^{-2i5 \cdot \arg(x_n - x_{n-1})}$$

$$= \Psi(x_1, \dots, x_N) \cdot e^{-2i5 \sum_{i=1} \arg(x_i - x_0)}$$

Exercise #12:

(pg 504)

(pg 504) "Auxiliary Integration"

$$\frac{1}{4\pi s} \int d^3x \phi \Delta \theta = \frac{1}{4\pi s} \int (\nabla \phi) \cdot (\nabla \theta) d^3x$$

$$+ \int_S \phi (\nabla \theta) \cdot \hat{n} dA$$

$$= \frac{1}{4\pi s} \int (\nabla \phi) \cdot (\nabla \theta) d^3x$$

(pg 496) "Abelian Chern-Simons Action"

$$S_{CS}[a] = \frac{k}{2\pi} \int d^3x a_z \partial_0 a_1$$

$$= \frac{1}{4\pi s} \cdot \frac{k}{2} \int d^3x a_z \partial_0 a_1$$

$$= \frac{1}{8\pi s} \int d^3x a_z \partial_0 a_1$$

Exercise #13:

(pg 507)

(8,93) "Action for effective low-energy"

$$S[A', a] = S_{CS, k}[A' + a] + S_{CS, k'}[a]$$

$$k = p$$

$$k' = \frac{1}{2s}$$

$$S[A', a] = \frac{k}{2\pi} \int d^3x (A' + a) \partial_0 (A' + a) + \frac{k'}{2\pi} \int d^3x a \partial_0 a$$

Problem 8.8.1 - Winding numbers:

a) (from problem) "Hamiltonian for particle on a ring"

$$\hat{H} = -\partial_\phi^2 / 2I$$

(from problem) "Quantum Partition Function"

$$Z = \text{tr} \cdot e^{-\beta \hat{H}}$$

$$\text{If } \psi = \cos(n\phi) + i \sin(n\phi)$$

$$\text{then } \frac{d\psi}{d\phi} = n(-\sin(n\phi) + i \cos(n\phi))$$

$$\begin{aligned} \frac{d^2\psi}{d\phi^2} &= -n^2(\cos(n\phi) + i \sin(n\phi)) \\ &= -n^2\psi \end{aligned}$$

Schrödinger equation: $\hat{H}\psi = E\psi$

$$-\frac{\partial^2}{2I}\psi = E\psi$$

$$-\frac{\partial^2}{2I}(\cos(n\phi) + i \sin(n\phi)) = \frac{n^2}{2I}\psi$$

$$E = \frac{n^2}{2I}$$

Partition Function:

$$\begin{aligned} Z &= \sum_{n=-\infty}^{\infty} e^{-\beta \hat{H}} \\ &= \sum_{n=-\infty}^{\infty} e^{-\beta \frac{n^2}{2I}} \end{aligned}$$

b) Partition function without trace:

$$\begin{aligned} Z &= \int_0^{2\pi} d\phi \langle \phi | e^{-\beta \hat{H}} | \phi \rangle \\ &= \int_0^{2\pi} d\phi \langle \phi | e^{-\beta (\frac{1}{2} I \dot{\phi}^2)} | \phi \rangle \\ &= \int_0^{2\pi} d\phi \int D\phi e^{-\beta (\frac{1}{2} I \dot{\phi}^2)} \\ &= \int_0^{2\pi} d\phi \int D\phi e^{-\frac{I}{2} \int_0^\beta d\tau \dot{\phi}^2} \end{aligned}$$

c) (from problem) "Classical Trajectories"

$$\phi(\tau) = \phi + 2\pi m \tau / \beta$$

(from problem) "General path"

$$\phi(\tau) = \bar{\phi}(\tau) + \eta(\tau)$$

where $\eta(\tau) = \text{"winding \#"}^{\circ}$

(from part b) "Quantum

Mechanical partition

function"

$$\begin{aligned}
 Z &= \int_0^{2\pi} d\phi \sum_{m=-\infty}^{\infty} \int D\phi e^{-\frac{I}{2} \int_0^{\beta} d\tau \dot{\phi}^2} \\
 &\quad - \frac{I}{2} \int_0^{\beta} d\tau (2\tau \bar{\phi}(\tau) + 2\pi m \tau / \beta) \\
 &\quad + \eta(\tau))^2 \\
 &= \int_0^{2\pi} d\phi \sum_{m=-\infty}^{\infty} \int D\phi e^{-\frac{I}{2} \left[\left(\frac{2\pi m}{\beta} \right)^2 \right]} \\
 &= Z_0 \sum_{m=-\infty}^{\infty} e^{-\frac{I}{2} \left[\left(\frac{2\pi m}{\beta} \right)^2 \right]}
 \end{aligned}$$

where $Z_0 = \int d\phi \int D\phi e^{-\frac{I}{2} \int_0^{\beta} K}$

$$\begin{aligned}
 K &= \dot{\phi} \dot{\eta} + \frac{2\pi \tau \eta}{\beta} + \eta \dot{\eta} + \eta \dot{\phi} \\
 &\quad + \frac{\pi \eta}{\beta} + \frac{2\pi \tau \dot{\phi}}{\beta} \\
 &\quad + \phi \dot{\phi} + \frac{2\pi \dot{\phi}}{\beta}
 \end{aligned}$$

So, $Z_0 = \int D\eta(t) e^{-\frac{I}{2} \int_0^{\beta} K}$

d) (from problem) "Poisson summation formula"

$$\sum_m h(m) = \sum_n \int_{-\infty}^{\infty} d\phi h(\phi) e^{2\pi i n \phi}$$

(8.96) "Partition function"

$$Z = Z_0 \sum_{m=-\infty}^{\infty} e^{-\frac{I}{2} \frac{(2\pi m)^2}{\beta}}$$

$$= Z_0 \int_{-\infty}^{\infty} dm e^{-\frac{I}{2} \frac{(2\pi m)^2}{\beta}}$$

$$= Z_0 \int_{-\infty}^{\infty} d\phi e^{-\frac{I}{2} \frac{(2\pi \phi)^2}{\beta}}$$

$$= Z_0 \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} d\phi e^{-\frac{I}{2} \frac{(2\pi \phi)^2}{\beta} + 2\pi i \phi}$$

$$= Z_0 \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} d\phi e^{\frac{(-2\pi i m)^2}{4 \left(\frac{I}{2} \frac{4\pi^2}{\beta}\right)}}$$

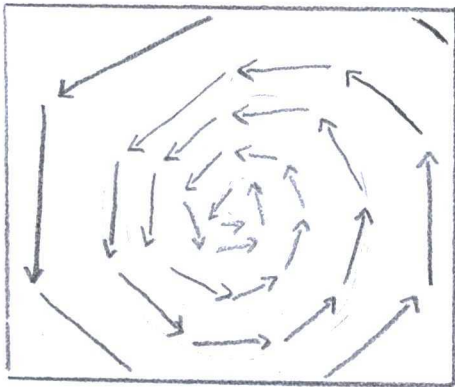
$$= Z_0 \sum_{n=-\infty}^{\infty} \sqrt{\frac{\beta}{2\pi I}} e^{-\frac{\beta}{2I} n^2}$$

Integral
Formula:

$$\int_{-\infty}^{\infty} e^{-(ax^2+bx)} dx = \sqrt{\frac{\pi}{a}} e^{\frac{b^2}{4a}}$$

Problem 8.8.2 - Topology of the magnetic

Skyrmion:



"Skyrmion"

a) (from problem) "Complex coordinate"

$$Z = x_1 + i x_2$$

(pg 444) "Topological action"

$$S_0[\phi] = \frac{1}{8\pi} \int d^2x \, \partial_\mu \mathbf{n} \cdot \partial_\mu \mathbf{n}$$

$$= \frac{3}{4\pi} \int dz d\bar{z} \left(\partial_z n_1 \partial_{\bar{z}} n_2 \pm \partial_{\bar{z}} n_1 \partial_z n_2 \right)$$

$$\boxed{d^2x = 6 dz d\bar{z}}$$

$$= \frac{3}{4\pi} \int dz d\bar{z} \, n_3 \left(\partial_z n_1 \partial_{\bar{z}} n_2 - \partial_{\bar{z}} n_1 \partial_z n_2 \right)$$

(from problem)

$$n_3 = \pm 1$$

b) (8.17) "Complex coordinate"

$$n_1 + i n_2 = \frac{2w}{1+|w|^2}, \quad n_3 = \frac{1-|w|^2}{1+|w|^2}$$

$$\text{Where } w = \prod_{i=1}^N \frac{z - a_i}{z - b_i}$$

$$S_{\text{Top}}[w] = -\frac{3i\theta}{2\pi} \int dz d\bar{z} \frac{1 - |w|^2}{1 + |w|^2} \left(\partial_z \frac{w}{1 + |w|^2} \right. \\ \left. \times \partial_{\bar{z}} \frac{w}{1 + |w|^2} - \partial_{\bar{z}} \frac{w}{1 + |w|^2} \partial_z \frac{w}{1 + |w|^2} \right)$$

$$\text{When } n_3 = \frac{1 - |w|^2}{1 + |w|^2}, \quad n_2 = -i \left(\frac{zw}{|w|^2 + 1} - a \right)$$

$$n_1 = \frac{zw}{|w|^2 + 1} - id$$

$$= \frac{3i\theta}{2\pi} \int dz d\bar{z} \frac{(1 - |w|^2)^2}{(1 + |w|^2)^4} \partial_z w \partial_{\bar{z}} \bar{w}$$

$$\Theta = -1$$

$$= iW\Theta \quad \text{.. if } W = \frac{3}{2\pi} \int dz d\bar{z} \frac{(1 - |w|^2)^2}{(1 + |w|^2)^4} \partial_z w \partial_{\bar{z}} \bar{w}$$

Problem 8.8.3- U(N) winding numbers:

a) (8.13) "Topological action"

$$d=1 \quad S_{\text{Top}}[g] = \frac{\Theta}{2\pi} \int dx \text{tr} (g^{-1} \partial_x g)$$

$$= \frac{\Theta}{2\pi} \int_0^1 dx \operatorname{tr} \left(e^{-ix} \frac{d}{dx} e^{ix} \right)$$

$$= \frac{\Theta}{2\pi} \int_0^1 dx \operatorname{tr} (e^{-ix} \cdot i e^{ix})$$

$$= i \mathbb{Z} \text{, when } \Theta = 2\pi \mathbb{Z} \text{ (pg 439)}$$

$\mathbb{Z}$ = integer
parameter

b) (9.14) "Topological action"

$$d=3, \quad S_{\text{top}}[g] = \frac{i\Theta}{24\pi^2} \int d^3x \epsilon^{ijk} \operatorname{tr}(\dots$$

$$\times (\bar{g}^{-1} \partial_i g) (\bar{g}^{-1} \partial_j g) (\bar{g}^{-1} \partial_k g))$$

$$\text{If } \Theta = 2\pi \mathbb{Z}$$

$$g = (e^{2\pi i X_i})^{n \cdot \sigma}$$

$$= (\cos(2\pi X_i) + i \sin(2\pi X_i))^{n \cdot \sigma}$$

$$\partial_x g = i \sin(2\pi X_i) \partial_i n(x_i) \cdot \sigma$$

$$S_{\text{top}}[g] = \frac{i}{12\pi} \int d^3x_i dx_j dx_k \epsilon^{ijk} ($$

$$\partial_i n(x_i) \cdot \sigma \partial_j n(x_j) \cdot \sigma \partial_k n(x_k) \cdot \sigma)$$

$$= 0$$

$$= 2??$$

c) (from problem) "Differential form"

$$\text{tr}(\bar{g}^{-1} \partial_x g) = \text{tr}(\bar{g}^{-1} dg \wedge g^{-1} dg \wedge g^{-1} dg)$$

$$\text{If } \text{tr}(\bar{g}^{-1} dg) = \phi$$

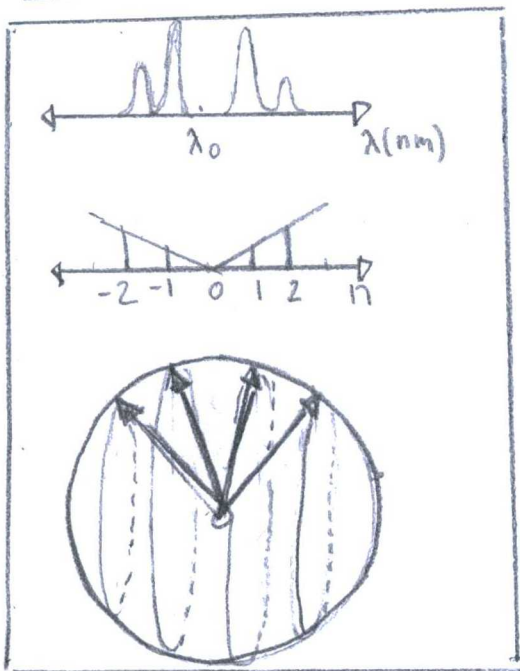
$$\text{and } \text{tr}(\hat{\phi} \wedge \hat{\phi}) = -\text{tr}(\hat{\phi} \wedge \hat{\phi})$$

$$\text{then } \text{tr}(\bar{g}^{-1} dg \wedge g^{-1} dg \wedge g^{-1} dg \wedge g^{-1} dg)$$

$$= \text{tr}(\hat{\phi} \wedge \hat{\phi} \wedge \hat{\phi} \wedge \hat{\phi})$$

$$= 0$$

Problem 8.8.4 - Wess-Zumino action from
Symmetry:



a) (from problem) "Fermion action"

$$S[\psi] = \int dx d\tau \sum_s \bar{\psi}_s (\partial_\tau - i \partial_x) \psi_s$$

(pg 486) "Complex coordinate"

$$Z = \frac{1}{\sqrt{2}} (\tau + iX) \quad \partial_Z = \frac{1}{\sqrt{2}} (\partial_\tau - i \partial_x)$$

(8.65) "Holomorphic currents"

$$j^{ab} = 2\pi^{\alpha\beta} 2\pi^{\alpha\beta} \quad \bar{j}^{ab} = 2\pi^{\alpha\beta} 2\pi^{\alpha\beta}$$

$$\partial_{\bar{z}} \bar{j}^{ab} = \partial_{\bar{z}} j^{ab} = 0$$

$$S[\pi] = \int dx d\tau \sum_s 2\pi_s (\partial_z \pi_s)$$

$$= 0 \quad \text{.. when} \quad \frac{j^{ab}}{2\pi} = 2\pi^{\alpha\beta}$$

b) (8.61) "Abelian Bosonization Action"

$$S_0[g] = \frac{1}{\lambda} \int dx d\tau \text{tr}(\partial_\mu g \partial_\mu g^{-1})$$

=

Integration by parts:

$$\int u dv = uv - \int v du$$

$$u = \partial_\mu g$$

$$du = \partial_\mu (\partial_\mu g)$$

$$v = g^{-1}$$

$$dv = \partial_\mu g^{-1}$$

$$= \frac{1}{\lambda} \left[(\partial_\mu g) g^{-1} - \int g^{-1} \partial_\mu (\partial_\mu g) \right]$$

c) (8.63) WZ term of $U(N)$

$$T[g] = -\frac{i}{12\pi} \int_B d^3x \epsilon^{ijk} \text{tr}(\tilde{g}^{-1} \partial_i \tilde{g} \tilde{g}^{-1} \partial_j \tilde{g} \tilde{g}^{-1} \partial_k \tilde{g})$$

$$= -\frac{i}{12\pi} \int_{\beta} \tilde{g}^* W$$

$$\text{where } \bar{w} = 3\text{tr}(\tilde{g}^{-1}\partial_i \tilde{g} \tilde{g}^{-1}\partial_j \tilde{g} \tilde{g}^{-1}\partial_k \tilde{g})$$

$$= 3\text{tr}(\tilde{g}^{-1}dW \tilde{g} \tilde{g}^{-1}dg)$$

$$(\text{from problem}) \quad g \rightarrow (1+W)g$$

$$T[Wg] = \frac{-i}{12\pi} \int d^3x \epsilon^{ijk} W \text{tr}(\tilde{g}^{-1}\partial_i \tilde{g} \tilde{g}^{-1}\partial_j \tilde{g} \tilde{g}^{-1}\partial_k \tilde{g})$$

$$= \frac{i}{4\pi} \int d^2x \epsilon_{\mu\nu} \text{tr}(W \partial_\mu g \partial_\nu g^{-1})$$

$$= T[(1+W)g] - T[g]$$

$$d) (\text{from part b}) \quad \partial_\mu g \partial_\mu g^{-1}$$

(0.66) "Bosonic representations of current"

$$j = \frac{1}{\sqrt{8\pi}} \tilde{g}^{-1} \partial_z g$$

$$\bar{j} = -\frac{1}{\sqrt{8\pi}} (\partial_z g) g^{-1}$$

$$S_0[g] + T[g] = \frac{1}{\lambda} \int dx d\tau \text{tr}(\partial_\mu g \partial_\mu g^{-1})$$

$$- \frac{i}{12\pi} \int d^3x \epsilon^{ijk} \text{tr}(\tilde{g}^{-1}\partial_i \tilde{g} \tilde{g}^{-1}\partial_j \tilde{g} \tilde{g}^{-1}\partial_k \tilde{g})$$

$$= \int dx d\tau \operatorname{tr}(\dot{j} \cdot g)^2$$

$$- i \frac{3\sqrt{2}\pi}{2} \int d^3x \epsilon^{ijk} \operatorname{tr}(j)^3$$

Problem 8.8.5 - Renormalization of the Wess-Zumino model:

(8.64) "Nonabelian Bosonization"

$$S[g] = S_0[g] + T[g]$$

(8.61) "Abelian Bosonization"

$$S_0[g] = \frac{1}{\lambda} \int dx d\tau \operatorname{tr}(\partial_\mu g \partial_\mu g^{-1}), \quad \lambda = 8\pi$$

$$\text{where } g = e^{z \cdot \theta}$$

(from problem) "Field splitting"

$$g = g_S g_F$$

(from problem) "Expansion"

$$g_F = 1 + W + \frac{1}{2} W^2 + \dots$$

$$\begin{aligned}
a) S[g_s g_F] &= \frac{1}{\lambda} \int d^4x d\tau \operatorname{tr}(\partial_\mu (g_s g_F) \partial_\mu (g_s g_F)^{-1}) \\
&= \frac{1}{\lambda} \int d^4x d\tau \operatorname{tr}(\partial_\mu (g_s [1 + W + \frac{1}{2} W^2]) \\
&\quad \times \partial_\mu (g_s [1 + W + \frac{1}{2} W^2])^{-1}) \\
&= \frac{1}{\lambda} \int d^4x d\tau \operatorname{tr}(\partial_\mu (g_s) \partial_\mu (g_s)^{-1}) \\
&\quad + \frac{1}{\lambda} \int d^4x d\tau \operatorname{tr}(\partial_\mu (W) \partial_\mu (W)^{-1}) \\
&\quad + \dots
\end{aligned}$$

(8.63) "WZ term of $U(N)$ "

$$\Gamma[g] = -\frac{i}{12\pi} \int_B d^3x \epsilon^{ijk} \operatorname{tr}(\tilde{g}^{-1} \partial_i \tilde{g} \tilde{g}^{-1} \partial_j \tilde{g} \tilde{g}^{-1} \partial_k \tilde{g})$$

$$\begin{aligned}
\Gamma[g_s g_F] &= -\frac{i}{12\pi} \int_B (g_s g_F)^* W \\
&= -\frac{i}{24\pi} \int d^4x \epsilon_{\mu\nu} \operatorname{tr}(g_s^{-1} \partial_\nu g_s) [W, \partial_\mu W]
\end{aligned}$$

$$S[g_s g_F] = S[g_s] + S^{(2)}[W] + S[g_s W] + \Gamma[g_s g_F]$$

$$\begin{aligned}
b) S[g_s, W] &= -\frac{i}{\lambda} \int d^4x \operatorname{tr} \left[(g_s^{-1} \partial_\mu g_s - \frac{i\lambda}{8\pi} \epsilon_{\mu\nu} g_s^{-1} \partial_\nu g_s) \right. \\
&\quad \left. \cdot [\partial_\mu W, W] \right]
\end{aligned}$$

$$= -\frac{1}{\lambda} \int d^2x \operatorname{tr} \left[(g_s^{-1} (\partial_\mu + \frac{i\lambda}{8\pi} \epsilon_{\mu\nu\rho} \partial_\nu) g_s) [\partial_\mu W_\rho W_\rho] \right]$$

$$= -\frac{1}{\lambda} \int d^2x \operatorname{tr} [\phi]$$

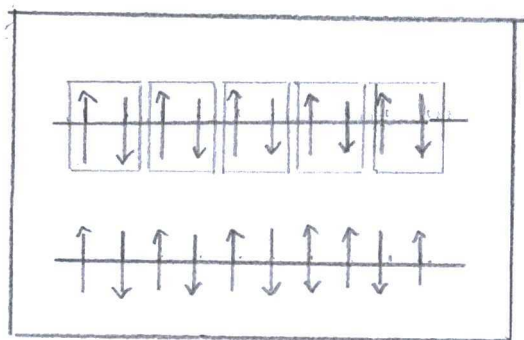
$$\text{When } \phi = g_s^{-1} (\partial_\mu + \frac{i\lambda}{8\pi} \epsilon_{\mu\nu\rho} \partial_\nu) g_s$$

$$\frac{1}{Z} \langle S[g_s, W]^2 \rangle_W = \frac{1}{2\lambda^2} \int d^2x \operatorname{tr} \phi_\mu \phi_\mu$$

$$\text{as } \lambda^2 = -\frac{4\pi}{N \ln b}$$

$$= -\frac{N \ln b}{8\pi} \int d^2x \operatorname{tr} (\phi_\mu \phi_\mu)$$

Problem 8.8.6 - Strong correlation physics via nonabelian bosonization:



"Spin chain
and
Néel phase"

(8.64) "Nonabelian Bosonization"

$$S[g] = S_0[g] + T[g]$$

(from problem) "Half-filled System"

$$\psi_e = e^{ik_F l} \psi_+(x_e) + e^{-ik_F l} \psi_-(x_e)$$

(from problem) "Lattice spin Operators"

$$S_e = \psi_e^{\alpha\dagger} \cdot \sigma^{\alpha\beta} \cdot \psi_e^\beta$$

$$= (e^{ik_{Fe}} \psi_+^{\alpha\dagger}(x_e) + e^{-ik_{Fe}} \psi_-^{\alpha\dagger}(x_e)) \cdot \sigma^{\alpha\beta} \cdot$$

$$\times (e^{ik_{Fe}} \psi_+^\beta(x_e) + e^{-ik_{Fe}} \psi_-^\beta(x_e))$$

$$= \psi_+^{\alpha\dagger}(x_e) \sigma^{\alpha\beta} \psi_+^\beta(x_e) + \psi_+^{\alpha\dagger}(x_e) \sigma^{\alpha\beta} \psi_-^\beta(x_e)$$

$$+ \psi_-^{\alpha\dagger}(x_e) \sigma^{\alpha\beta} \psi_+^\beta(x_e) + \psi_-^{\alpha\dagger}(x_e) \sigma^{\alpha\beta} \psi_-^\beta(x_e)$$

(0.66) "Bosonic representation
of the current"

$$j = \frac{1}{\sqrt{3}\pi} \bar{g} \partial_z g \quad \bar{j} = -\frac{1}{\sqrt{3}\pi} (\partial_z g) g^{-1}$$

(0.67) "General Nonabelian prefactor"

$$\psi_+^{\alpha\dagger} \psi_-^b \longrightarrow \gamma_g^{ab}$$

The inessential c and c' constants
from the problem.

$$S_e = \gamma_{g_{++}}^{\alpha b} \sigma^{\alpha\beta} + \gamma_{g_+}^{ab} \sigma^{\alpha b} + \gamma_{g_{-+}}^{ab} \sigma^{\alpha\beta} \\ + \gamma_{g_{--}}^{\alpha\beta} \sigma^{\alpha\beta}$$

$$= \text{tr} [c(\dot{g}_+^s + \dot{g}_-^s) + c'(-)(\dot{g} + \dot{g}')] \sigma]$$

$$\text{When } c = \frac{\sqrt{g} \pi}{\dot{g} \partial_z g} = \frac{\sqrt{g} \pi}{\dot{g} \partial_{\bar{z}} g}$$

$$c' = \frac{\sqrt{g} \pi}{\dot{g}^{-1} \partial_z g} = \frac{\sqrt{g} \pi}{\dot{g}^{-1} \partial_{\bar{z}} g}$$

b) (from problem) "Translationally invariant massive action"

$$S_m[g] \equiv m \int \text{tr}(g^2 + g^{-2})$$

The partition function ($Z = \int D\psi e^{-S[\psi]}$)

on page 455 requires multiples of 2π in exponent for "spin chains,"

A positive mass-coefficient needs n multiples of π in $\text{tr}(g^2 + g^{-2})$. While negative mass, a value would give zero exponent in the action.

c) Effective action:

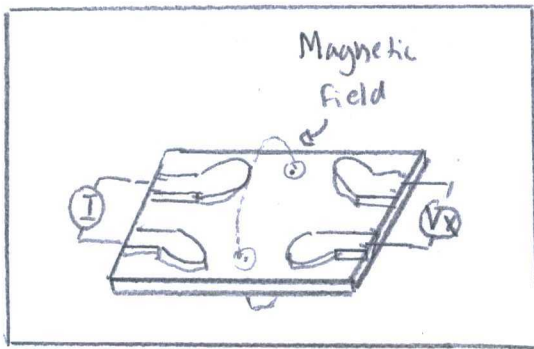
(from problem) $g \rightarrow i n \cdot \sigma$

(from problem) "Gradient term"

$$\begin{aligned} n_c S_0 [i n \cdot \sigma] &= n_c \cdot \frac{1}{\lambda} \int dx d\tau \text{tr}(\partial_\mu g \partial_\mu \bar{g}) \\ &= \frac{n_c}{\lambda} \int dx d\tau \text{tr}(\partial_\mu i n \cdot \sigma (\partial_\mu i n \cdot \sigma)) \end{aligned}$$

$$\begin{aligned} n_c T [i n \cdot \sigma] &= n_c \cdot \frac{-i}{12\pi} \int d^3x e^{i\theta} \text{tr}(g^{-1} \partial_\mu g g^{-1} \partial_\nu g g^{-1} \partial_\mu g) \\ &= n_c \cdot \frac{i\theta}{24\pi^2} \int d^3x \text{tr}(g^{-1} \partial_\mu g g^{-1} \partial_\nu g g^{-1} \partial_\mu g) \end{aligned}$$

Problem 8.8.7 - Knot invariant from Chern-Simons functional:



"Two-dimensional
abelian anyon braiding"

(8.78) "Gauge Potential Coupling"

$$S_{CS}[a] \rightarrow S_{CS}[a] + \int d^3x j^\mu a_\mu$$

(from problem) "Lorentz gauge"

$$\partial^\mu a'_\mu = \partial_t a'_0 - \partial_i a'_i = 0$$

a) If $X'_\mu = (\tau, \mathbf{x})$ and $\alpha'(a_0, \mathbf{a})$

then $\partial^\mu a_\mu = \partial_\tau a_0 - \partial_\epsilon a_\epsilon$

$$= -i \partial_\tau a_0' - \partial_\epsilon a_0$$

$$= -i \partial_\epsilon a_0' + i \partial_\tau a_0$$

$$= 0$$

Current Vector Potentials

$$S_{cs}[a] + \underbrace{\int d^3x j^\mu a_\mu}_{\text{"Current Vector potential"}}$$

"Current Vector potential"

If $j = \partial_\tau X_\mu' \delta(\mathbf{x} - \mathbf{x}_\mu(\tau))$

then $S_{cs}[a] = iE \int d^3x d\tau \frac{j^\mu a_\mu}{X}$

b) If $a_{\mu, q^0} a_{\nu, q} = -i \frac{2\pi}{k} \epsilon_{\mu\nu\rho} \frac{q_\rho}{q^2} \delta q_\mu - q_\nu$

and $\partial_0 a_1 = a$

(from problem) "Additional term"

$$\frac{T}{2} \int d^3x (\partial_\mu a_\mu)^2$$

(pg 496) "Abelian Chern-Simons Action"

$$S_{CS}[a] = \frac{k}{2\pi} \int d^3x a_2 \partial_0 a_1$$

(p. 78) "Gauge Potential Coupling"

$$S_{CS}[a] \longrightarrow S_{CS}[a] + \int d^3x j^\mu a_\mu$$

$$S_{CS}[a] = -i \int \epsilon_{\mu\nu\sigma} \frac{q_\sigma}{q^2} + \frac{\Gamma}{2} \int d^3x (a_\mu a_\mu)$$

$$= \frac{1}{2} \sum_q a_{\mu q} (-i \epsilon_{\mu\nu\sigma} q_\sigma + \Gamma q_\mu q_\nu) a_{\nu, -q}$$

$$= 0 \text{ from } q_\mu a_{\mu, q} = 0$$

$$Z = \int D\phi e^{-S_{CS}[a] + \int d^3x j^\mu a_\mu}$$

$$= \int D\phi e^{-\int d^3x j^\mu a_\mu}$$

c) (from problem)

$$A = e^{-\frac{i\pi}{k} \sum \epsilon_{\mu\nu\sigma} j_{\mu, q} \frac{q_\sigma}{q^2} j_{\nu, -q}}$$

$$= e^{-\frac{i\pi}{k} \sum j_{\mu, q} \cdot b_\mu} \quad \text{... if } b_\mu = \epsilon_{\mu\nu\sigma} \frac{q_\sigma}{q^2}$$

$$= e^{-\frac{i\pi}{k} \sum \int d\tau \frac{dx}{d\tau} \cdot b_\mu}$$

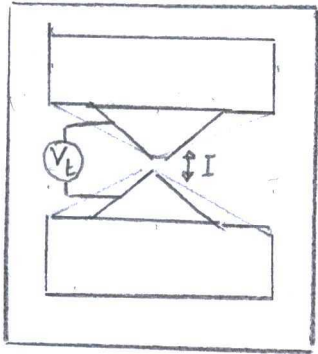
$$= e^{-\frac{i\pi}{k} \sum \int ds (\nabla \cdot b_\mu)} \quad \text{... if } \frac{dx}{d\tau} = j$$

$$= e^{-\frac{2i\pi}{k} I}$$

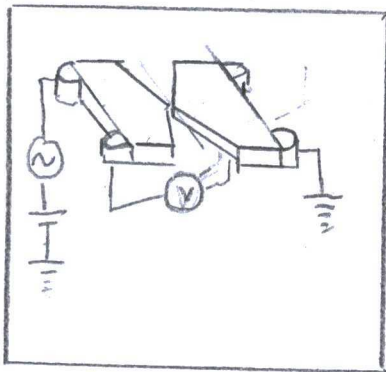
$$I = \int ds (\nabla \cdot \mathbf{b})$$

Problem 8.8.8 - Fractional Quantum Hall Effect:

physics at the edge



"Fractional Quantum Hall effect"



"Hall Resonance"

$$(8.36) \rho = \frac{\text{voltage}}{\text{current}} \\ = \frac{h}{e^2 \nu}$$

Cite: Wen, Xiao-Gang

"Theory of the edge states
in fractional Quantum Hall
effects"

Cambridge, MA (1992)

Cite: Cohen, Liam

"Universal Chiral Luttinger
liquid behavior in a
graphene fractional
Quantum Hall point contact"

Santa Barbara, CA (2023)

a) Energy cost of a boundary condition:

(from problem) "Boundary Potential"

$$V = E X_L$$

$$H_1 = \int dx dx_L \rho v$$

$$= E \int dx \int_0^{h(x)} dx_L X_L$$

$$= E \int dx \rho(x)$$

(in problem) "charge density"

$$\rho(x) dx = \sigma h(x) dx$$

Equation of motion:

("Continuity equation")

$$\partial_t \rho - v \partial_x \rho = 0$$

$$v = \frac{\partial_t \rho}{\partial_x \rho}$$

$$= \frac{E}{B}$$

Effective action:

(from problem) "New variable"

$$\partial_x \phi = \frac{2\pi}{v} \rho$$

(8.101) "Effective action"

$$S[\phi] = \frac{v}{4\pi} \int dx d\tau [v(\partial_x \phi)^2 - \partial_x \phi \partial_t \phi]$$

The "new variable" relates to

the continuity equation because

$$\partial_t \phi \partial_x \phi - v(\partial_x^2 \phi) = 0$$

b) (8.101) "Effective action"

$$S[\phi] = \frac{v}{4\pi} \int dx d\tau [v(\partial_x \phi)^2 - \partial_x \phi \partial_t \phi]$$

Gauge: $\partial_\mu \phi \rightarrow \partial_\mu \phi - A_\mu$

$$S[\phi] = \frac{v}{4\pi} \int dx d\tau \left[(v(\partial_x \phi - A_x)^2 - (\partial_x \phi - A_x) \cdot (\partial_t \phi - A_t) + \partial_t \phi A_x - \partial_x \phi A_t) \right]$$